JAMES MITCHELL

Research Engineer

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PROFESSIONAL SUMMARY

I'm a Research Engineer with 6 years of experience building scalable systems and investigating emerging technologies in machine learning and distributed computing. Based in Glasgow, I've worked across fintech, healthcare, and infrastructure sectors, focusing on translating research concepts into production systems. I'm particularly interested in optimization algorithms and their real-world applications, and I enjoy mentoring junior team members through complex technical challenges.

WORK EXPERIENCE

Senior Research Engineer

Nexus Labs

nexuslabs.io

Glasgow, United Kingdom

March 2022 - Present

Led the development of a distributed parameter optimization framework that reduced model training time by 40% across five concurrent projects and established performance benchmarking standards for the research team. Designed and implemented automated testing infrastructure using pytest and continuous integration pipelines that caught 85% of regression issues before deployment. Mentored three junior engineers in Python best practices and research methodology, with two subsequently promoted to mid-level positions. Published two peer-reviewed papers on resource allocation in heterogeneous computing environments with collaboration from academic partners.

Research Engineer

Quantum Insights Ltd

quantuminsights.co.uk

Edinburgh, United Kingdom

July 2019 - February 2022

Developed statistical analysis tools in Python and R for processing financial time series data, enabling the risk analysis team to reduce computation time from 8 hours to 45 minutes per analysis cycle. Implemented containerized microservices using Docker and Kubernetes that improved deployment consistency and reduced production incidents by 60%. Collaborated with external research institutions to validate novel anomaly detection algorithms against real-world datasets with 92% precision rates. Built comprehensive documentation and runbooks that became the foundation for knowledge transfer during team expansion.

Junior Research Engineer

DataFlow Systems

dataflow.net

Glasgow, United Kingdom

January 2018 - June 2019

Prototyped and evaluated machine learning models for healthcare data analysis, achieving 87% accuracy on patient outcome prediction using ensemble methods. Contributed to codebase maintenance and performed code

reviews for four senior engineers, identifying performance bottlenecks that improved query response times. Assisted in designing experiment tracking systems using MLflow that standardized reproducibility across research projects. Troubleshot infrastructure issues and provided technical support to stakeholders during system deployments.

EDUCATION

Doctor of Philosophy in Computer Science

University of Edinburgh

Edinburgh, United Kingdom

Graduated: June 2017

Bachelor of Science in Computer Science with Mathematics

University of Glasgow

Glasgow, United Kingdom

Graduated: June 2015

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Bash, R

Machine Learning: TensorFlow, PyTorch, scikit-learn, ensemble methods, statistical analysis

Data Engineering: Apache Spark, Kafka, distributed systems, ETL pipelines

Tools and Platforms: Docker, Kubernetes, Jenkins, Git, Linux, PostgreSQL, MongoDB

Research Methods: Experimental design, A/B testing, statistical validation, academic publishing

Cloud Services: AWS, Google Cloud Platform basic familiarity

ADDITIONAL INFORMATION

Publications: 3 peer-reviewed conference papers on optimization and distributed computing

Certifications: AWS Certified Cloud Practitioner

Languages: English (native), French (conversational)